GlucoTwin

A virtual patient workspace for glucose forecasting

Historical clinical context + wearable sensor history

Research prototype | 11 October 2026

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Problem and healthcare use case

A clinician needs to see how recent glucose changes relate to a patient's historical clinical context.

GlucoTwin links the profile to visible sensor history and fixed future forecasts.

The research objective is advance prediction of a new sustained high-glucose episode.

Linked historical and sensor data

CGMacros historical input: age, recorded sex, BMI, baseline HbA1c and fasting glucose.

Dynamic input: CGM glucose and heart rate, linked to the same patient.

Retrospective replay; CGM samples are reconstructed and clinical availability is assumed.

Fusion inside the model

Both streams enter the same fitted feature vector; the profile is more than display context.

CGMacros uses 5 historical fields, 22 CGM features and 9 heart-rate features.

Imputation and scaling use fitting patients only. Fusion's predictive advantage remains uncertain.

Runtime and research architecture

Historical profile + available sensor history -> causal features -> saved fused Ridge -> local Python API -> browser replay.

Separate saved XGBoost path -> uncalibrated onset score and frozen cutoff -> research decision ledger with delayed outcomes.

DCLP3 is an offline 30/60-minute numerical study. Event replay uses excluded evaluation folds; operational alerts remain disabled.

Clinician interaction demonstration

Inspect historical profile, sensor coverage and visible glucose history.

Advance the replay; forecasts stay fixed when issued. Outcomes appear only when available.

Compare model error with persistence, switch patients and examine missing observations.

CGMacros event research: fixed score/cutoff/decision; outcome hidden until issue time +45 minutes.

Advance-event definition

New sustained high onset: 4 consecutive five-minute readings above 180 mg/dL, after 7 readings at or below 180.

Prediction window: onset 5-30 minutes ahead. Features contain only past information.

27 development patients; 3 outer folds separate fitting, cutoff setting and evaluation. Six reserved patients remain unscored.

Measured event prediction

Metric	Fused XGBoost v2
Episodes detected	229 / 336 (68.2%)
Conservative precision	52.8%
False alarms / eligible day	0.98
Median detected-event lead	10 minutes

Development results. Cutoff-setting folds were not consistently successful; patient intervals cross gate limits.

Larger DCLP3 numerical study

Participant MAE (mg/dL)	30 minutes	60 minutes
Persistence	20.92	32.32
CGM-only XGBoost	17.53	27.47

134 development patients, 4 patient-disjoint folds; 34 reserved unscored. Separate cohort; approximately 35/65-minute lookahead from the latest reading.

Limits that affect the conclusion

Only 36.6% of evaluable episodes received at least ten minutes warning in the v2 development backtest.

Later onset v3 recall fell to 51.5%; it was rejected as an overall improvement.

DCLP3 low-target 60-minute MAE: 54.31 versus persistence 50.04. Clinical fields added no convincing overall gain.

No clinical validation. Risk percentages, operational alerts and treatment advice remain disabled.

Reproducibility and remaining work

Python + scikit-learn + XGBoost; HTML/CSS/JavaScript browser. Measured studies ran on local CPU.

Reproducible scripts, frozen configurations and aggregate reports are in the repository; records and model binaries stay local.

Next: freeze a final model/cutoff policy before using reserved data. Research replay does not establish clinical reliability.

Original code: MIT License. Dataset terms remain separate. Video preparation is deferred.